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IDs: 202301812, 202300056, 202301826, 202300741, 202300886

Date: May 2, 2025



MATH205

Report for project on the *Collatz conjecture*.

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1 Introduction

1.1 Definition and background information

The *Collatz conjecture*, also known as the *$3x+1$ conjecture*, is a simple mathematical problem that has challenged mathematicians for years. Despite its rather simple definition, no one has yet been able to prove or disprove it conclusively.

The conjecture states that, starting with any positive integer x , repeatedly applying the function $C(x)$ defined below will eventually lead to 1:

$$C(x) = \begin{cases} 3x + 1 & \text{if } x \equiv 1 \pmod{2}, \\ \frac{x}{2} & \text{if } x \equiv 0 \pmod{2}. \end{cases}$$

That means, if the number is odd, multiply it by 3 and add 1; if it is even, divide it by 2. This process is repeated recursively. Although this algorithm appears straightforward, the long-term behavior of all starting values remains unsolved.

This project explores the background, behavior, attempted proof strategies, and applications related to the conjecture.

2 Main Body

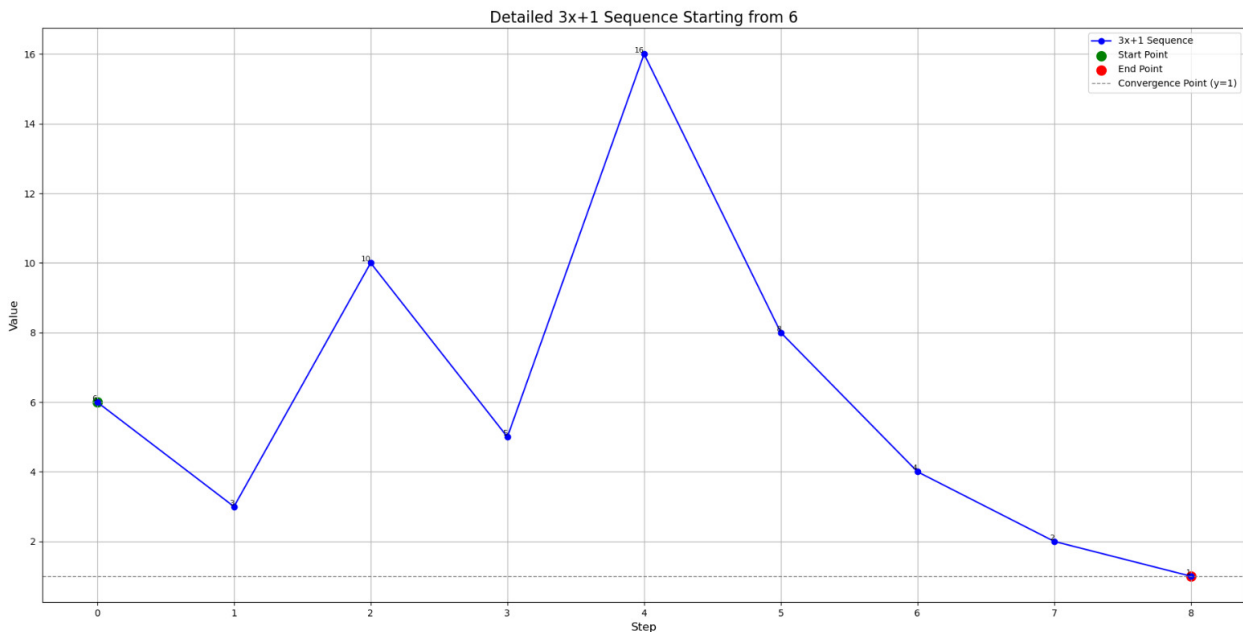
2.1 Detailed information on the conjecture

The Collatz conjecture is deceptively simple in formulation but deeply complex in behavior. It concerns the iterative behavior of integers under a piecewise-defined function:

$$C(x) = \begin{cases} 3x + 1 & \text{if } x \equiv 1 \pmod{2}, \\ \frac{x}{2} & \text{if } x \equiv 0 \pmod{2}. \end{cases}$$

This process, often visualized as a sequence starting from a positive integer x , is repeated recursively. For instance, starting at $x = 6$, the sequence evolves as:

$6 \rightarrow 3 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$



The conjecture asserts that no matter which positive integer is chosen initially, the sequence always eventually reaches the cycle $4 \rightarrow 2 \rightarrow 1$, after which it loops indefinitely.

Despite extensive numerical verification up to extremely large bounds (as in the work of Oliveira e Silva), no formal proof or counterexample exists. The problem's appeal lies in its accessibility: it requires no advanced mathematics to understand, yet its resolution remains elusive even to experts.

For example, Various generalizations and analogues of the problem, such as the $px + q$ variants, have been proposed to explore similar dynamics in broader contexts.

2.2 Relevance and Applications

Even though the Collatz conjecture hasn't been solved yet, its recursive structure and behavior puzzled mathematicians and computer scientists alike. It's used in understanding algorithmic predictability limits, is frequently mentioned in complexity theory and dynamical systems, for example the growth rates, stopping times, and total stopping times of sequences have been subjects of rigorous study.[1]

Moreover, due to its simplicity, it's used as a teaching example in courses that cover topics such as algorithms, number theory, and discrete mathematics.[1]

In addition, The Collatz conjecture is not only a mathematical puzzle but also a practical benchmark for computational systems. Its iterative complexity and unpredictable behavior make it ideal for testing algorithmic efficiency and system performance. Researchers like Oliveira e Silva have used it to push the limits of parallel and distributed computing, making the conjecture a tool for evaluating large-scale computational capabilities.[4]

2.3 Attempted proof

We attempt to prove the Collatz conjecture by contradiction. Suppose there exists an infinite sequence Q that never reaches 1. Then $1 \notin Q$, and

$$Q = \{Q_1, Q_2, Q_3, \dots\}$$

with $f_c(Q_i) = Q_{i+1}$, where

$$f_c(x) = \begin{cases} 3x + 1 & \text{if } x \equiv 1 \pmod{2}, \\ \frac{x}{2} & \text{if } x \equiv 0 \pmod{2}. \end{cases}$$

We sort the sequence Q in ascending order—justified by the Axiom of Choice [1]—and denote it as Q' . Then define:

$$Q_E = \{E_1, E_2, \dots\}, \quad Q_O = \{O_1, O_2, \dots\}$$

with:

$$E_i < E_{i+1}, \quad O_i < O_{i+1}$$

Even-to-Odd Mapping. If $f_c(E_1) = O_x$ for some $x > 1$, then $O_x < O_1$, contradicting the ordering. Therefore, $f_c(E_1) = O_1$. By the same logic:

$$f_c(E_r) = O_r \quad \text{for all } r \in \mathbb{N}$$

Odd-to-Even Mapping. Assume $f_c(O_1) = 3O_1 + 1 = 2^r \cdot O_y$ for $r \geq 2$. Then $O_y < O_1$, contradicting the ordering. So:

$$f_c(O_r) = E_{r+1} \quad \text{for all } r \in \mathbb{N}$$

Hence:

$$Q' = \{E_1, O_1, E_2, O_2, E_3, O_3, \dots\}$$

and since Q and Q' contain the same elements, they are equal sets [1]:

$$Q = Q'$$

Contradiction. Let an odd number be $O_x = 2^k m - 1$, with m odd and $k \geq 1$. Then repeated application of f_c yields:

$$\begin{array}{ll} 2^4 \cdot 5 - 1 & (\text{odd}) \\ \rightarrow 3 \cdot 2^4 \cdot 5 - 2 & (\text{even}) \\ \rightarrow 3 \cdot 2^3 \cdot 5 - 1 & (\text{odd}) \\ \rightarrow 3^2 \cdot 2^3 \cdot 5 - 2 & (\text{even}) \\ \dots & \\ \rightarrow 3^4 \cdot 5 - 1 & (\text{even}) \end{array}$$

This shows at least two consecutive even numbers (E-E) appear, contradicting the alternating pattern $E, O, E, O, \dots$ established in Q .

Conclusion. The pattern generated by f_c contradicts the assumed structure of the non-terminating Q . Therefore, no such infinite sequence can exist. This supports the Collatz conjecture [1].

2.4 Algorithm that applies the Collatz conjecture with its analysis

This is a basic algorithm that applies the Collatz conjecture:

```
# Collatz.py
def IsEven(n):
    if (n % 2 == 0):
        return True
    else:
        return False

def Collatz(x, sols):
    if x == 1:
        sols.append(1)
        return 1
    sols.append(x)
    if IsEven(x):
        return Collatz(x // 2, sols)
    else:
        return Collatz(3 * x + 1, sols)
```

Here is the Algorithm's analysis:

The base case for the algorithm is: $T(1) = 1$

The recurrence relation for the algorithm is:

$$T(x) = \begin{cases} T\left(\frac{x}{2}\right) + 1 & \text{if } x \text{ is even,} \\ T(3x + 1) + 1 & \text{if } x \text{ is odd.} \end{cases}$$

Time complexity: The time complexity of the algorithm is unknown because the Collatz conjecture remains unproven. Empirically, the algorithm terminates for all tested inputs, but the number of steps required to reach 1 varies significantly depending on the input.

Space complexity: The space complexity of the algorithm is $O(n)$, where n is the number of steps in the sequence, because the algorithm stores the sequence in a list.

3 Conclusion

The Collatz conjecture remains unresolved. While it has no known practical applications, its simplicity and unpredictability make it a useful illustration in algorithm design, number theory, and computational benchmarking [4]. The conjecture continues to captivate mathematicians as a striking example of how a basic problem can resist resolution for decades.

References

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